MOHAMMED ABDUL RAHMAN

GenAI Engineer | RAG & Agentic AI | AWS Bedrock | Production LLM Systems | Healthcare Automation

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PROFESSIONAL SUMMARY

Generative AI Engineer with ~2 years of experience building and shipping **production-grade LLM systems** for enterprise healthcare automation. Proven track record in **RAG, Graph-RAG, and Agentic AI architectures** on AWS Bedrock. Delivered measurable business impact — processing **500–1,200 medical charts/day**, cutting processing time from 30 min to under 2 min (**15x throughput**), enabling ~**25% workload reduction** across 1,000+ coders. Full-cycle ownership from system architecture to cloud deployment on AWS.

TECH STACK

Languages	Python · SQL · Bash
Agentic AI	Agentic Workflows · Multi-Agent Systems · Autonomous Agents · Model Context Protocol (MCP)
Generative AI & LLMs	RAG · Graph-RAG · Prompt Engineering · LLM Evaluation & Optimization · Fine-tuning
AI Frameworks	LangChain · LangGraph · CrewAI · FastAPI · Flask · Agno
Vector & Graph DBs	FAISS · ChromaDB · Pinecone · Neo4j
MLOps	MLflow · LangSmith · Docker · GitHub Actions · CI/CD Pipelines
Cloud & Infra	AWS Bedrock · AWS SageMaker · AWS S3 · AWS Lambda · Azure · GCP · Linux · GitLab
NLP & Deep Learning	Transformers (BERT, GPT) · NER · Text Classification · Sentiment Analysis · CNN · LSTM · TensorFlow · PyTorch · Scikit-learn
Data & Visualization	Pandas · NumPy · Power BI · EDA · Feature Engineering · Statistical Analysis

PROFESSIONAL EXPERIENCE

Ascent Healthcare Pvt Ltd | AI/ML Engineer — *May 2025 – Present*

- Built a production **LLM-powered medical coding system on Amazon Bedrock** converting unstructured clinical notes into ICD-10, CPT, and modifier codes processing **500–1,200 charts/day** at **75% ICD, 86% CPT, and 70% modifier accuracy**.
- Achieved **15x throughput improvement** (30 min → under 2 min per chart), enabling an estimated **~25% reduction in manual workload** across **1,000+ medical coders**, directly cutting operational costs.
- Architected a **Graph-RAG pipeline** combining Neo4j knowledge graphs with semantic vector retrieval (Pinecone, ChromaDB) for context-aware extraction of complex medical coding knowledge from clinical text.
- Deployed an **Agentic AI knowledge assistant** on internal SOPs and manuals, adopted across departments, onboarding **~1,000 new joiners** with sub-second query latency, cutting policy lookup time by **~60%**, handling **500+ requests/day** at peak.
- Implemented **HIPAA-compliant RAG architecture** on AWS (Lambda, S3, EC2) with advanced prompt engineering and LangChain-based workflows for clinical entity extraction and billing code generation.
- Managed end-to-end system lifecycle: data ingestion pipelines, model experimentation with **LangSmith**, deployment workflows, and production monitoring. Collaborated with SMEs and data engineers to resolve edge cases and align outputs with compliance requirements.

TheSmartBridge Pvt Ltd | Machine Learning Engineer (Intern) — *Aug 2024 – Mar 2025*

Operated at full-time engineer capacity · Independently owned product delivery cycles across RAG, MLOps, and deployment

- Owned end-to-end architecture of **RAG pipelines** for enterprise document analysis — independently designed and delivered production-ready systems for client-facing edtech products serving **thousands of learners**.
- Designed and deployed **LLM-based document summarization and QA systems** that improved response relevance and task accuracy for client-facing edtech products.

- Reduced deployment turnaround by **~40%** by re-engineering ML training pipelines and introducing **MLflow-based experiment tracking** — workflow adopted as the team standard.
- Established **CI/CD pipelines using Docker and GitHub Actions** for automated model deployment and monitoring, bringing DevOps discipline to a previously manual process.
- Built and productionised ML pipelines (ANN, CNN) for classification tasks across structured and image data, managing full lifecycle from preprocessing to **REST API deployment**.
- Mentored junior engineers on RAG architecture patterns and MLflow best practices, directly contributing to team capability uplift.

LetsGrowMore | **Data Science Engineer** *May 2024 – Aug 2024* Remote

- Processed and analyzed **100+ datasets** using Pandas and NumPy, delivering EDA reports and feature engineering pipelines across diverse business domains.
- Built and evaluated end-to-end ML classification and regression models using Scikit-learn, covering preprocessing, training, and performance evaluation.
- Designed **interactive Power BI dashboards** translating complex data outputs into actionable business intelligence for internal stakeholders.

KEY PROJECTS

Graph-RAG Medical Coding System | *AWS Bedrock · Neo4j · LangChain · LangGraph*

- Production system processing **500–1,200 clinical charts/day**, converting unstructured notes to ICD-10/CPT codes at **75–86% accuracy** with **15x throughput improvement** over manual coding. HIPAA-compliant deployment on AWS.

Agentic Knowledge Assistant | *LangGraph · CrewAI · FAISS · FastAPI*

- Multi-agent system onboarding **~1,000 new joiners** with sub-second response latency and **500+ requests/day** at peak — reduced policy lookup time by **~60%** for healthcare operations teams.

Enterprise RAG Document Intelligence Pipeline | *LangChain · ChromaDB · FastAPI · MLflow*

- End-to-end RAG pipeline for multi-document QA and summarization deployed as a REST API. MLflow experiment tracking reduced deployment turnaround by **40%** and was adopted team-wide at Smart Bridge.

Chess Piece Detection — Deep Learning | *DenseNet121 · TensorFlow · OpenCV*

- Built a real-time detection system for 12 chess piece classes using DenseNet121 with data augmentation and advanced preprocessing, achieving high-accuracy classification for real-time recognition.

CERTIFICATIONS

✦ AWS Certified Cloud Practitioner — Amazon Web Services	✦ TCS ION Career Edge — Young Professional
✦ MySQL RDBMS Certification	✦ Business Analytics with Excel
✦ Power BI Certification	✦ AWS ML Specialty (Pursuing)

EDUCATION

Master of Business Administration (MBA) — IT Specialisation *Dec 2021 – Oct 2023*

Osmania University, Hyderabad | Relevant coursework: SQL, Power BI, Business Analytics

Bachelor's in Commerce & Computer Applications *Jun 2018 – Sep 2021*

Osmania University, Hyderabad

ADDITIONAL INFORMATION

Languages: English · Hindi · Telugu · Urdu (Professional Working Proficiency)

Soft Skills: Agile Project Management · Technical Documentation · Cross-functional Collaboration · Problem Solving